

# Intergenerational Housing Misallocation and Fertility

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**Abstract**

# 1 Introduction

## 2 Empirical Evidence

Families occupy larger homes and are more likely to own them. In the 2023 American Housing Survey, 81.3 percent of owner-occupied homes have at least three bedrooms, compared with 30.2 percent of renter-occupied homes. Among homes with four or more bedrooms, 9.2 percent are rented. These figures describe the allocation of the housing stock. They motivate a housing market in which size and tenure interact, without establishing that large rentals are unavailable to an otherwise identical household.

The Panel Study of Income Dynamics (PSID) provides a longitudinal view of housing around childbirth. I date first birth from a woman's recorded biological-child history and compare her housing at observed interviews before and after that event. The current estimate for the change in rooms between event year  $-1$  and event year  $+3$  is 0.72 rooms, with a woman-clustered standard error of 0.09. The comparison uses women with no recorded children as controls, and includes individual and survey-year effects. Housing may change before a birth, so this event pattern is a description of family housing trajectories rather than the effect of an unanticipated child. The biennial interview schedule leaves some birth cohorts without an observed reference period, so the magnitude remains provisional.

The American Community Survey (ACS) supplies a complementary comparison on a larger cross-section. Following a matched pseudo-panel design, women observed with children are paired with childless donors who have the same observed demographic characteristics in earlier survey years. The donors supply a synthetic prebirth housing path; the mothers supply observed housing after the inferred event. The Northeast implementation estimates separate curves for rooms, bedrooms, and ownership, and a restricted event window checks common cohort support. This exercise asks whether the broad housing pattern survives in a different data source. Its prebirth observations are constructed matches, and the ACS roster does not reveal a complete birth history.

These facts discipline different parts of the quantitative analysis. The tenure-by-size distribution describes the housing bundles households occupy; the PSID and ACS trajectories describe housing around entry into motherhood. Neither an occupied stock share nor a matched event curve isolates an exogenous shift in fertility. Appendix ?? defines the samples, housing measures, event clocks, and limits of the exploratory birth-instrument analyses.

## 3 Model

In this Section, I develop a model of fertility and housing decisions. Agents live a finite number of periods in an overlapping generations economy. They are subject to persistent idiosyncratic income shock and transitory fertility and housing taste shock. They derive utility from consumption, housing, and fertility, and can save using a one-period liquid asset or by purchasing a home. They can either rent or own a home, and pay transaction costs when seeling the home. Owners can borrow against their home equity, subject to liquidity constraint. Households' have a fertile period of their lives during which they can attempt to have kids, which give them utility but affect their housing and consumption needs. They also have a retirement age, after which they do not work and receive pension benefits. Such benefits are financed by a payroll tax which is imposed on workers, who supply labor inelastically.

### 3.1 The economy

I begin by describing agents and their preferences, the income process, the markets in which agents can trade and the fertility process. Time is discrete.

**Households and Preferences.** The economy is populated by overlapping generation of households, who live for at most  $T$  periods. Households are also denoted as ‘adults’. Adult age is denoted by  $a \in [0, J]$ . Agents can die at each period, and I denote  $s_a$  their probability of surviving to next period. Households work before retirement age  $a_R$  and can attempt births during the fertile window  $a \in [a_F^0, a_F^1]$ . Let  $n_t$  denote children ever born and  $m_t$  children currently at home. Children are dependents and do not work. Each dependent child matures and leaves the household with a fixed probability  $\text{fix}$  each period, independently of the other children. Maturation reduces  $m_t$  but does not change  $n_t$ . Households value consumption, housing services, and children at home, and discount future utility by  $\beta$ . At any period  $t$ , agents have flow utility

$$u_t(c_t, s_t; m_t) = \frac{\mathcal{C}(c_t, s_t; m_t)^{1-\sigma}}{1-\sigma} + \psi_t m_t.$$

The parameter  $\sigma > 0$  governs curvature over the consumption aggregator  $\mathcal{C}$ , with the logarithmic limit understood when  $\sigma = 1$ , while  $\psi_t$  measures the direct utility benefit of a child at home. For a given number of children,  $\mathcal{C}$  is increasing and concave in consumption and housing services. Children increase the household’s needs, so  $\mathcal{C}_m < 0$ , and raise the marginal contribution of housing services to the aggregator,  $\mathcal{C}_{sm} > 0$ . The functional form of  $\mathcal{C}$  is specified in the quantification.

Households either rent or own their residence. An owner occupies  $h_t$  units of housing, while a renter occupies  $h_t^R$  units. Housing services are

$$s_t = \mathbf{1}\{\text{own}\}\chi h_t + \mathbf{1}\{\text{rent}\}h_t^R, \quad \chi > 1. \quad (1)$$

The parameter  $\chi$  captures the additional services households obtain from owning a home of a given size.

At death, households value the estate they leave. Let  $w_{t+1}^e$  denote bequeathable wealth, comprising liquid wealth and housing value. Bequest utility is

$$B(w_{t+1}^e) = \theta_0 \frac{(\theta_1 + w_{t+1}^e)^{1-\sigma}}{1-\sigma}. \quad (2)$$

The parameter  $\theta_0$  scales the strength of the bequest motive, while  $\theta_1$  shifts the wealth level at which it is evaluated; the logarithmic limit again applies when  $\sigma = 1$ .

**Fertility.** During fertile ages, a household chooses whether to attempt a birth. An attempt succeeds with age-specific probability  $\pi_a$  and, upon success, increases both  $n_t$  and  $m_t$  by one. The first birth carries a one-time utility cost  $\xi$ . Households observe extreme-value taste shocks  $\varepsilon_t^F$  before choosing whether to attempt; their scale is  $\kappa_1$  for the first birth and  $\kappa_C$  for subsequent births. Birth success is then realized. Households subsequently observe housing taste shocks  $\varepsilon_t^H$  and choose housing and saving conditional on the realized number of children. Thus, housing can adjust to a birth within the period.

**Production and income.** Competitive firms produce the consumption good using effective labor  $L_t$  with constant productivity  $A$ :

$$Y_t = AL_t, \quad w = A. \quad (3)$$

The wage per unit of effective labor therefore equals labor productivity. A working household supplies  $e_a z_t$  units of effective labor, where  $e_a$  is the age-earnings profile and  $z_t$  exposes the household to idiosyncratic earnings risk. Earnings are taxed at rate  $\tau^w$ . Retirees receive a pay-as-you-go pension  $\varpi_t$ , financed by current payroll taxes. Disposable income is

$$y_t^d(a, z_t) = \begin{cases} (1 - \tau^w) w e_a z_t, & a < a_R, \\ \varpi_t, & a \geq a_R. \end{cases} \quad (4)$$

Households also receive a lump-sum transfer  $T_t$  from property-tax revenue, which is rebated equally across households.

**Housing markets.** A unit of housing trades at price  $P_t$ . Aggregate housing supply is

$$H_t^s = H_0 P_t^\eta. \quad (5)$$

Here  $H_0$  sets the scale of supply and  $\eta$  is its price elasticity. Owned housing can be as large as  $\bar{h}$ , whereas rental housing is limited to  $\bar{h}^R < \bar{h}$ . The largest homes are therefore available only through ownership. Housing depreciates at rate  $\delta_H$  and is subject to a property tax at rate  $\tau_t^p$ . Selling an owner-occupied home costs a fraction  $\psi^s$  of its sale value.

Competitive landlords rent housing at a unit rent  $r_t$ . A landlord purchases at  $P_t$ , pays depreciation and property tax, and resells at  $P_{t+1}$ . Equating the return to the gross bond return  $R_b$  gives

$$r_t = (R_b + \delta_H + \tau_t^p) P_t - P_{t+1}. \quad (6)$$

Rent covers financing, depreciation, and property taxes, net of the change in the value of the house.

**Saving and mortgage finance.** Households save in a one-period bond with gross return  $R_b$ . Borrowing is secured by housing: an owner can borrow up to a fraction  $\phi$  of the home's purchase value, while renters cannot borrow. Buying a home requires a down payment of  $(1 - \phi) P_t h_t$ . The household must finance this payment out of beginning-of-period liquid wealth and the proceeds from selling its previous home, net of selling costs. Current earnings and transfers arrive after the housing transaction and cannot finance the down payment. Households that retain their home do not make a new down payment.

## 4 Quantification

I calibrate a stationary reference economy representing 2007, before the subsequent decline in fertility. I use observations from the surrounding years to discipline fertility, housing, and household wealth. Housing moments use the national household sample in the 2005–2006 ACS; fertility moments come from the 2004 and 2006 CPS and 2003–2006 natality records, while wealth moments use the 2003 and 2005 PSID. This initial steady state provides the starting point for the historical transition, in which changes in fertility preferences account for the subsequent path of births while the other structural parameters remain fixed.<sup>1</sup>

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The

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<sup>1</sup>This working section describes the 2007 calibration under review. The persistent-earnings process and parenthood housing floor provide the comparison specification; earnings, entry-wealth mapping, and the alternative utility with child-dependent housing shares remain provisional. No new set of parameter estimates or model fit is adopted here.

second consists of parameters estimated internally by matching model moments to their empirical counterparts. I describe the two groups in turn, explain which moments are particularly informative about the internally estimated parameters, and then assess the model’s fit.

## 4.1 Parameters Set Outside the Estimation

I first describe the parameters assigned outside the estimation. Table 1 summarizes their values and restrictions.

**Demographics and preferences.** A model period is four years. Households enter adult life at age 18, retire at age 66, and live at most through age 85. Survival is certain before retirement and thereafter follows age-specific survival probabilities. Dependent children remain in the household for an expected 18 years, implying a maturity probability of  $2/9$  per period. I set risk aversion to  $\sigma = 2$  and retain the power equivalence scale with child weight  $\omega = 0.7$  and exponent  $\nu = 0.7$ . The consumption weight is  $\alpha_0 = 0.733$ , measured using childless renters in the 2019–2023 CEX. These preference restrictions are held fixed in the estimation; they need not be measured in the base year.

**Housing finance and transaction costs.** I set the annual real return on liquid wealth to  $i_{\text{ann}} = 0.02$ , the proportional selling cost to  $\psi^s = 0.06$ , and the labor-income tax rate to  $\tau_{\text{pay}} = 0.179$ , retaining the external restrictions used in the housing-finance calibration (Greaney et al., 2025). Annual housing depreciation is  $\delta_{\text{ann}} = 0.011$  and the annual property-tax rate is  $\tau_{\text{ann}}^p = 0.01$ . Mortgage debt cannot exceed a share  $\phi = 0.80$  of the value of the house, so a purchase requires at least 20 percent equity. Purchase eligibility accounts for current earnings as well as existing resources. The model imposes no separate payment-to-income limit. Payroll-tax revenue finances retirement benefits through the stationary pension budget; the pension is therefore determined within each candidate equilibrium.

**Income process.** I impose a deterministic lifecycle profile of household earnings,  $e_a$ , normalized to average one over working ages. Within age, log earnings follow a persistent AR(1) process estimated from the PSID at the model’s four-year frequency. The working process has persistence  $\rho_z = 0.7346$  and innovation standard deviation  $\sigma_z = 0.4838$  in log earnings. It has no fixed permanent household types and is approximated with 15 income states. Entrants draw from the stationary earnings-risk distribution.<sup>2</sup>

**Housing products and supply.** Housing is indivisible and available in a limited assortment of types and sizes (Davis and Van Nieuwerburgh, 2015). I represent this lumpiness with five owner-occupied house sizes,  $\{H_k\}_{k=1}^5 = \{2, 4, 6, 8, 10\}$  rooms. Renters choose housing continuously up to  $\bar{h}^R = 6$  rooms. These menus summarize the tenure-specific distribution of rooms. I retain a housing-supply elasticity of  $\eta = 0.63$ ; the supply intercept  $H_0$  is estimated jointly with household parameters.

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<sup>2</sup>The comparison with a persistent process plus an independent transitory component is retained as an alternative. Its four-year persistence is 0.7761, persistent innovation standard deviation is 0.4374, and transitory standard deviation is 0.1650. Neither adding transitory risk nor assuming stationary income risk at age 18 is an adopted resolution of the earnings specification.

**Entry wealth and bequests.** Households enter at age 18 with heterogeneous nonhousing wealth and debt, constructed from the PSID wealth-to-family-income distribution of childless renters aged 18–24. The joint entry distribution retains the reference mapping between wealth and earnings.<sup>3</sup> Bequest utility does not vary directly with the number of children. Children affect estate accumulation through their effects on housing needs, tenure, and saving.

Table 1: Parameters set outside the estimation

| Parameter                         | Value          | Evidence or restriction           |
|-----------------------------------|----------------|-----------------------------------|
| Model period                      | 4 years        | Model timing                      |
| Entry, retirement, maximum ages   | 18, 66, 85     | Lifecycle specification           |
| Risk aversion $\sigma$            | 2              | Maintained preference restriction |
| Equivalence scale $(\omega, \nu)$ | (0.7, 0.7)     | Maintained power scale            |
| Consumption weight $\alpha_0$     | 0.733          | CEX 2019–2023, childless renters  |
| Annual real return                | 2%             | External return restriction       |
| Annual depreciation; property tax | 1.1%; 1%       | Maintained housing costs          |
| Financed share $\phi$             | 0.80           | 20% equity requirement            |
| Sale cost $\psi^s$                | 6%             | External transaction cost         |
| Payroll tax                       | 17.9%          | External tax restriction          |
| Income persistence $\rho_z$       | 0.7346         | Four-year PSID process            |
| Income innovation SD $\sigma_z$   | 0.4838         | Four-year log-earnings innovation |
| Owner house sizes                 | 2, 4, 6, 8, 10 | Rooms                             |
| Largest rental unit               | 6              | Rooms                             |
| Housing-supply elasticity $\eta$  | 0.63           | Maintained supply restriction     |
| Tenure taste-shock scale          | 0.005          | Fixed in the working calibration  |
| Entry wealth and debt             | Heterogeneous  | PSID childless renters aged 18–24 |

## 4.2 Internally Estimated Parameters

I estimate nine parameters jointly using twelve moments describing childlessness, fertility timing and family size, housing consumption, tenure, and wealth accumulation. Separately, I choose the fertility utility scale  $\psi_0$  so that mean children ever born among households aged 46 and above is 2.1 in the stationary economy. This is a normalization of the initial steady state, not a period total fertility rate. Although the parameters are chosen jointly, I organize the discussion into economic blocks and describe which moments are most informative about each block.

**Saving.** The annual discount factor  $\beta_{\text{annual}}$  governs households’ willingness to carry resources across ages. Aggregate wealth relative to annual gross earnings is particularly informative about it. The empirical target is a ratio of aggregate wealth to aggregate earnings, rather than the average of household wealth-to-earnings ratios.

**Housing demand, tenure, and supply.** I use housing-size differences by family composition to discipline housing demand. The room response around the first birth is particularly informative about the parenthood housing requirement  $h_P$ . The difference between larger and smaller families provides additional information about housing demand under the maintained equivalence scale.

<sup>3</sup>The empirical family-income denominator must be reconciled with the model’s gross-earnings units. The working experiments retain the inherited heterogeneous entry law; zero entrant wealth is not imposed in this calibration.

The floor is the same for all households with dependent children, rather than increasing with each additional child. Ownership rates and the ownership gap associated with recent parenthood are particularly informative about the owner housing-service multiplier  $\chi$ , while mean occupied rooms helps discipline the housing-supply intercept  $H_0$ .<sup>4</sup>

**Fertility and family size.** Childlessness is particularly informative about the first-birth utility cost  $\xi$ . Mean age at first birth and the share of first births at age 30 or older help discipline first-birth taste dispersion  $\kappa_1$ . The fraction of mothers with exactly one child helps discipline later-birth taste dispersion  $\kappa_C$ . These parameters are estimated jointly because the same family outcomes respond to the costs, timing, and utility value of children.

**Bequests.** The bequest parameters  $\theta_0$  and  $\theta_1$  govern the strength and wealth dependence of the bequest motive. Annual bequests relative to aggregate wealth are particularly informative about  $\theta_0$ , while the p90-to-median ratio of wealth relative to family income among households aged 76–84 provides information about the wealth shift  $\theta_1$ . These moments also respond to saving and tenure choices, so the two bequest parameters are disciplined jointly with those blocks.

Table 2: Internally calibrated parameters and informative moments

| Parameter               | Interpretation                   | Particularly informative moments                   |
|-------------------------|----------------------------------|----------------------------------------------------|
| $\beta_{\text{annual}}$ | Annual discount factor           | Wealth / gross earnings                            |
| $\xi$                   | First-birth utility cost         | Childlessness                                      |
| $\kappa_1$              | First-birth taste dispersion     | Mean first-birth age; share at 30+                 |
| $\kappa_C$              | Later-birth taste dispersion     | Exactly one child among mothers                    |
| $h_P$                   | Parenthood housing floor         | First-birth room response; family room differences |
| $\chi$                  | Owner housing-service multiplier | Ownership level; recent-parent gap                 |
| $H_0$                   | Housing-supply intercept         | Mean occupied rooms                                |
| $\theta_0$              | Bequest-motive strength          | Annual bequests / wealth                           |
| $\theta_1$              | Bequest wealth shift             | Old-age wealth/income p90 / median                 |
| $\psi_0$                | Utility from children            | Mean children ever born: 2.1 normalization         |

### 4.3 Target Moments and Fit

The target system combines children-ever-born distributions from the CPS, first-birth timing from natality records, tenure and housing-size moments from the ACS, housing responses around childbirth from the PSID event study, and wealth moments from the PSID. Table 3 reports the working targets for the 2007 reference economy.<sup>5</sup>

<sup>4</sup>The alternative utility removes  $h_P$  and estimates two child-dependent housing-share parameters, a parenthood shift and a slope in the number of dependent children. That specification has ten jointly searched parameters and the same separate fertility normalization.

<sup>5</sup>The empirical first-birth room response and recent-parent ownership gap require a common observation rule in the data and model before their fit can be interpreted. The other population, timing, and weighting choices are being reviewed row by row. Model moments and parameter estimates will be inserted for the chosen specification; historical July fits and unadopted experimental fits are not substituted for that comparison.

Table 3: Calibration targets for the 2007 steady state

| Moment                                            | Target | Model |
|---------------------------------------------------|--------|-------|
| Mean children ever born, ages 46+ (normalization) | 2.10   | —     |
| Childlessness, women 40–44 (%)                    | 19.83  | —     |
| Exactly one child among mothers, 40–44 (%)        | 21.37  | —     |
| Mean age at first birth (years)                   | 25.98  | —     |
| First births at age 30+ (%)                       | 24.93  | —     |
| Mean occupied rooms, capped at nine               | 5.608  | —     |
| Ownership, household heads 30–55 (%)              | 67.63  | —     |
| First-birth room response, $-1$ to $+3$ (rooms)   | 0.720  | —     |
| Rooms: 3+ versus 1–2 resident children            | 0.385  | —     |
| Recent-parent ownership gap (percentage points)   | 12.76  | —     |
| Wealth / annual gross earnings                    | 6.146  | —     |
| Annual bequests / wealth (%)                      | 0.88   | —     |
| Wealth/income p90 / median, ages 76–84            | 3.516  | —     |

## 5 Policy

## 6 Conclusion

## References

- Morris A. Davis and Stijn Van Nieuwerburgh. Housing, Finance, and the Macroeconomy. In *Handbook of Regional and Urban Economics*, volume 5, pages 753–811. Elsevier, 2015. ISBN 978-0-444-59533-1. doi: 10.1016/B978-0-444-59531-7.00012-0.
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